

SPECIFICATIONS

EXTENDED LENGTH RAISED ROOF ADA CONVERSION VAN WITH WHEELCHAIR LIFT

*****Note: All specifications shall be considered minimum unless stated otherwise*****

EXCEPTIONS TO THESE REQUIREMENTS AND SPECIFICATIONS: If a bidder has any exceptions to these requirements, such exceptions must be stated in writing and describe what is proposed to be furnished in lieu of the specified requirements.

SECTION 1 – CHASSIS

1.1 DIMENSIONS:

- a. Wheel base: 147"
- b. Overall height: 100"
- c. Inside height: 67"

NOTE: these Dimensions apply to completed van conversion.

1.2 COOLING SYSTEM: Heavy duty radiator with capacity available to prevent overheating while operating in stop and go transit operation in ambient temperatures as high as 110 degrees F. and provide anti-freeze protection to -30 degrees F required.

1.3 ENGINE AND TRANSMISSION:

- a. Engine: Provide minimum 300 hp at 5000 rpm and 390ft. lb. of torque 225 rpm.
- b. Transmission: 5 speed automatic with overdrive, minimum.

1.4 STEERING: Power, with easily identifiable reservoir.

1.5 GWVR: 9,000 LBS.

1.6 AXLES:

- a. Front: To meet GVWR
- b. Rear: To Meet GVWR
- c. Gear Ratio: must provide minimum of 65 mph.
- d. Alignment: Successful bidder is responsible for aligning the front axle to within chassis mfr's standards before delivery and provides a copy of the alignment report with the delivery vehicle.

1.7 ELECTRICAL:

- a. Battery: 12 volt, 650 CCA, maintenance free type.
- b. Alternator: 200 AMP

1.8 BRAKES: Power assisted, ABS, Heavy-Duty hydraulic, self-adjusting disc or drum type.

1.9 DRIVESHAFT:

- a. The driveshaft shall be rated and capable of transmitting the torque multiplication of the engine/transmission to the drive wheels. The chassis OEM shall balance the drive shaft.
- b. Drive shaft guard: to prevent drive shaft from hitting the ground, drive shaft guard shall be sized to allow for movement of the U-Joint assemblies.

SECTION 2 – BODY**2.1 GENERAL:**

- a. If modified, vehicle to be reinforced such that structure of the basic van is not degraded.
- b. Completed vehicle shall meet all applicable requirements of the AMERICANS WITH DISABILITIES ACT (As set forth in CFR 37 & 38, issued 9.6.1991, with respect to the body structures).

2.2 RAISED ROOF: If not OEM the following specification must be met.

- a. Roof Lines: Shall be arched and present a symmetrical contour.
- b. Roof Joint: The raised roof shall be completely joined to become an integral part of the basic van. Both outer and inner roof shall be attached to the body in the same manner to prevent leakage. The new top must be completely sealed with an anti-fungal sealant.
- c. Insulation: 1" fiberglass blanket, polystyrene bead board or equivalent to prevent heat loss in cold weather and cool air in hot weather. Interior of the roof must be covered with an OEM style headliner with a minimum 6 A/C vents.
- d. Color: The exterior color of the raised roof shall be white and match the white of the chassis. The color may be "Gel Coat" or painted, free of brush or swirl marks.
- e. Steel Rollover Cage: Roof shall contain a collapse-resistant steel rollover cage certified to meet FMVSS 220 standards.
- f. A certified copy of the FMVSS 220 test results for the type vehicle to be provided must be included with the bid to be considered for award.

2.3 WINDOWS:

- a. Tint: Chassis standard windshield and front most doors on the basic van be retained and shall have mfr's tint.
- b. Glass: Window shall be OEM safety glass and uniformly tinted.
- c. Safety Standards: All windows including windshield and tinting shall meet all applicable federal and state motor vehicle safety standards.

2.4 AUDIBLE BACK-UP WARNING DEVICE: Required, shall sound when in reverse.**2.5 FINISHING PROCEDURES: If not OEM the following specification must be met**

- a. Bare metal components - Must be prepped with Ditzler DAX1980 primer sealer or equal and finished with Ditzler Acrylic Enamel paint or equal to match chassis
- b. Compatibility - primer and finish must be compatible with one another.
- c. Welded Areas - In addition to A & B above welded areas must be weld slag and impurities.

2.6 UNDERCOATING:

- a. The entire underside of the vehicle body, including the undersides of fenders, shall be coated with a fire resistant asphalt base, rubber base, or equal.
- b. Application of undercoating by conversation mfr shall not jeopardize corrosion protection warranty issued by the chassis mfr.

SECTION 3- INTERIOR**3.1 GENERAL INTERIOR:**

- a. Interior color shall be color-coordinated and complimentary to the van's exterior color.
- b. All sharp edges and sharp corners shall be eliminated; any fastenings or other objects that can catch a passenger's clothing or that can cause injury shall not be permitted. No abrasions, marks, or cuts will be acceptable on any of the interior walls or seats.

3.2 PANELING OR TRIM:

- a. Interior trim package shall be OEM. Aftermarket interiors are not acceptable.
- b. All rivets, screws, snaps, etc. in paneling shall be counter sunk, pan head, or be designed in such a manner that they do not snag clothing or skin as the vehicle occupants move about in the van. All interior panels, materials, and treatments shall meet all FMVSS 302 requirements.

3.3 INSULATION:

- a. The vehicle body shall be fully insulated in the roof and all body panels, including all extended top and bottom door panels to deaden sound and reduce vibrations and heat transfers.
- b. A minimum 1-1/2" thick blanket of fiberglass or other fire resistant insulation material for the OEM van heavy duty insulation shall be provided in sidewalls.

3.4 FLOORING:

- a. The subflooring shall be a minimum of 3/4" thick plywood securely fastened to the understructure. All overcuts and indentions shall be filled, all edges, cutouts, and/or notches shall be properly sealed. The subfloor shall be sanded smooth to ensure a proper bonding of the vinyl flooring.
- b. A grey transit vehicle vinyl floor material will be applied with a minimum thickness of 2.2 millimeters. Examples: Altro or Gerfloor
- c. There shall be no bubbles or blisters in the floor covering.

3.5 FLOOR PLAN & SEATING CAPACITY:

- a. Capacity for:
 - a.1 Driver, 5 ambulatory and 2 wheelchairs or,
 - a.2 Driver, 7 ambulatory and 1 wheelchair or,
 - a.3 Driver and 9 ambulatory when not carrying a wheelchair.
- b. See floor plan in Section 7 for details on placement.
- c. Passenger seats shall be arranged such that the unobstructed hip-to-knee room, measured at seat-level for each seated passenger, shall not be less than 26".

3.6 PASSENGER RESTRAINT SYSTEM:

- a. Shoulder restraints systems and retractors shall be provided on seats as required by applicable FMVSS regulations. On in-board forward facing seats, child safety seats anchors meeting FMVSS 225 will be provided.
- b. Seat belts shall be integrated into seating provided or securely attached to OEM locations provided by the vehicle mfr.

3.7 HEATING AND COOLING:

- a. Dash unit shall be separately controlled from auxiliary system.
- b. An OEM auxiliary rear heater/air conditioning system with the maximum output capacity available from the chassis mfr shall be provided.

3.8 DRIVER'S AND PASSENGERS STEPS:

- a. Driver's step shall be a running board (step) made of galvanized expanded steel and be a minimum 9" wide by 40" long. The step top should be between 11" and 12" from the ground. It will be mounted securely with a minimum of 4 bolts and a stabilizing arm. It shall incorporate a galvanized steel mud flap.
- b. A passenger side running board (step) shall be provided and be of the same construction as the driver's side step. It shall be a minimum 9" wide by 85" long and extend rearward from the front wheel well. The step top should be between 11" and 12" from the ground. It will be mounted securely with a minimum of 6 bolts and 2 stabilizing arms. It shall support 500 lbs. at the midpoint.

SECTION 4 – ADDITIONAL REQUIREMENTS

- 4.1 **BACKUP CAMERA:** Shall offer a backup camera that will have the camera mounted above the rear window, with a minimum 4 inch video screen mounted in the driver's area for ease of viewing but does not interfere with driver's view. Shall be a heavy duty system capable of enduring a transit environment and be either mounted in a recessed area or have a guard mounted to prevent damage to camera.

- 4.2 **SURVEILLANCE SYSTEM:** Shall be equipped with a multi-camera or dual-vision surveillance system to provide continuous digital video recording from the exterior forward perspective as well as interior activities of the driver and passengers. Surveillance system shall include installation.

4.3 EMERGENCY AND SAFETY EQUIPMENT:

- a. A rechargeable 5 lbs dry chemical fire extinguisher with and ABC rating shall be provided and installed in each van. The fire extinguisher shall have a gauge showing the status of the charge.
- b. Three folding reflective triangles compliant with federal DOT standard 125 shall be provided with each van. The triangle shall be stored in a sturdy non-cardboard case.
- c. OEM tire changing tools & jack shall be provided and mounted.

- d. In addition to these specifications, all related equipment shall be designed, built, attached and operated in accordance with all applicable safety codes and design standards including the following:
 - d.1 Society of automotive engineers (electrical components and wiring, hydraulic components, fasteners)
 - d.2 American National Standards Institute (Chain drive and wire rope components)
 - d.3 American Welding Society (Welding code and recommended practices)
 - d.4 Federal Motor Vehicle Safety Standards (FMVSS)
- e. All parts shall be new. All necessary servicing and adjustments shall be made on the equipment prior to delivery of the vehicle. All exposed metal surfaces shall be painted or shall be corrosion-resistant.

SECTION 5 - TRAINING & MANUALS

- 5.1 **TRAINING:** The successful vendor shall provide each grantee/agency receiving a conversion van with one (1) hour of operator training on the safety use, care and maintenance of any pertinent aspects of the van.
- 5.2 **MANUALS:** The successful vendor shall provide with each van a maintenance manual or one maintenance CD. The maintenance manual shall detail preventive maintenance procedures, troubleshooting and electrical diagrams of the lift system.

SECTION 6 - WARRANTY:

- 6.1 **REQUIREMENTS:** Warranties are in addition to any statutory, implied warranties, remedies or warranties imposed by the contractor. Consistent with this requirement that the contractor warrants and guarantees that the contractor warrants and guarantees that each van will be free of defects for one year or 25,000 miles, whichever comes first. The contractor shall also warrant and guarantee that fiberglass roof and roll cage will be free of defects for three years or 50,000 miles, whichever comes first, and that the wheelchair lift will be free of defects for two years or 25,000 miles, whichever comes first. This warranty covers only the system installed by the contractor and not the entire vehicle.
- 6.2 **VOIDING THE WARRANTY:** The warranty will not apply to any part or component installed that has been subject to misuse, negligence, or accident, or that has been repaired or altered in any way insofar as such repairs were in accordance with the contractor's supplied maintenance manuals.
- 6.3 **WARRANTY REPAIRS:**
 - a. The contractor is responsible for all repairs during the warranty period to the equipment installed. Once notified of warranty a matter by the recipient, the contractor will have 48 hours to arrange the needed repairs. If the contractor fails to start the repairs within 48 hours, the van recipient may have repairs completed

- by a third party shop and bill the contractor at the shop hourly rate plus the cost of any materials plus handling fee of 15% on all materials used.
- b. If a component fails during the warranty period, the warranty on the repaired or replaced equipment is equal to original warranty at the time that the installation was complete.

SECTION 7-CONVERSION SPECIFICATIONS:

If not completely OEM the following specifications must be met.

- 7.1 CONVERSION/MODIFIER REQUIREMENTS: Vehicle modifier or conversion company must be certified by the chassis mfr. to perform the requested work per this RFB. Example: Ford QVM. Converter must be listed by the FTA as a qualified TVM program manufacturer.
- 7.2 SERVICE FACILITY LOCATIONS: Successful vendor must have a warranty service providers for chassis and body located within the State of Tennessee.
- 7.3 ELECTRICAL:
 - a. Wiring: SXL copper strand, gauge to meet or exceed the required current load.
 - b. All non-chassis OEM wiring shall be color coded & numbered.
 - c. All non OEM wiring shall be loomed.
 - d. Clips shall be rubber or plastic coated to prevent cutting thru the wiring insulation.
 - e. Protective grommets shall be installed at where wiring passes through metal.
 - f. Where wires are extended, no lock wire connectors will be allowed. Insulated stake-on spade terminals or equal shall be used.
- 7.4 BODY: If not completely OEM the following specification must be met.
 - a. Raised roof: OEM roof shall be removed and replaced with a reverse flange or similar type fiberglass 16" to 18" aerodynamic raised roof with a 60" minimum center aisle height. Roof shall be painted to match van. Gel coat only is not acceptable.
 - b. Side entry doors: Swing open chassis van doors shall be retained. Chassis side entry doors shall be modified to increase minimum entry height to 57" and facilitate safe boarding. Door top extensions shall be galvanized steel or composite construction. To prevent water/air leaks, door top extensions should maintain OEM radius at corners. All doors shall be easily operable from within.
- 7.5 ASSIST POLE: A padded stainless steel stanchion bar shall be installed at the side entrance door. The padded bar shall be safety yellow in color. Stanchion shall be mounted to the floor and to the ceiling in a structural member.
- 7.6 SEATING AND INTERIOR ITEMS: If possible all OEM seating should be used. If aftermarket seating is used the following specifications apply:
 - a. Driver's and co-pilot seats - chassis mfr's high back bucket seats.

- b. All seats rearward of the b pillar shall be forward facing. Each seat position shall be a minimum of 16" wide and 17" deep 3pt seat. Example: Freedman GOes fixed or forward facing folding or equal.
- c. All seats: A minimum of 26" hip to knee room shall be provided. Grey 36 oz. Transportation grade vinyl. All metal components shall be powder coated and capable of meeting the following tests:
 - c.1 Salt spray 1000 hrs.
 - c.2 ASTM DL 17 humidity resistance 1000 hrs.
 - c.3 ASTM D2247 Impact resistance 1000 hrs.
 - c.4 ASTM D2794

7.7 WHEELCHAIR LIFT: Braun Century II or equal with installation meeting ADA and NHTSA.

7.8 WHEELCHAIR TIE DOWNS:

- a. Q-Straint QRT Deluxe or equal with 3 pt occupant restraints and L track anchorage.
- b. Flush L track minimum 36" wide for rear anchorage. L track or L pods for front anchorage spaced a minimum 36" wide.
- c. Front wheelchair position 48" long and rear wheelchair position minimum 52" long.
- d. The installation of these securement systems must be tested in the chassis being proposed (i.e. Ford Transit chassis) to withstand a force to each securement strap of 2,500 lbs. in a forward longitudinal direction and 5,000 lbs. for each mobility aid.

7.9 INTERNATIONAL SIGN FOR ACCESSIBILITY: Two adhesive backed 14' X 14" international signs for accessibility shall be shipped loose inside the van.

7.10 INTERIOR LIGHTING:

- a. Vehicle interior shall be illuminated by a minimum of 3 L.E.D. light fixtures.
- b. Lighting fixtures shall be controlled by the chassis OEM lighting switch.
- c. Opening any door shall automatically illuminate interior as long as the doors are open.
- d. Two (2) L.E.D. side passenger door stepwell lights shall be provided.

7.11 PRIORITY SEATING SIGNS: Vehicle shall contain a sign which indicates that seats in the front of the vehicle are priority seats for persons with disabilities.

7.12 FLOOR PLAN:

